



HK IMO Training 2019-2020
Problem Set 3
For 19 October 2019



Problem 9. Let ABC be a triangle with $\angle BAC > 90^\circ$. Let D be a point on the segment BC and E be a point on the line AD such that AB is tangent to the circumcircle of $\triangle ACD$ at A and BE is perpendicular to AD . Given that $CA = CD$ and $AE = CE$, determine $\angle BCA$.

Problem 10. Find all positive integers n such that

$$\frac{n^{3n-2} - 3n + 1}{3n - 2}$$

is an integer.

Problem 11. The incircle of an acute-angled triangle ABC with centre I touches sides BC and AB at points A' and C' respectively. Let M be the midpoint of side AC , and let N be the midpoint of arc AC , containing B , in the circumcircle of $\triangle ABC$. If P is the projection of the point M to the segment $A'C'$, prove that the points I , P , and N are collinear.

Problem 12. In a country, there are 2018 cities, some of which are connected by roads. Each city is connected to at least three other cities. It is possible to travel from any city to any other city using one or more roads. For each pair of cities, consider the shortest route between these two cities. What is the greatest number of roads that can be on such a shortest route?